Mean Reversion + Beta-Zero Targeting

A Long/Short Equity Strategy by
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Summary

Research supports the idea that a mean reversion strategy, particularly with a near-zero beta, can be successful. These findings were examined by creating and backtesting an algorithm that would trade autonomously while following a set of instructions with optimized parameters. With the results from the backtest and live paper trading, it can be concluded that the strategy could plausibly perform well, but some improvements should be made to make it more robust.

Motivation and Rationale

The idea of creating an algorithm that would trade on its own and perform well is interesting both personally and professionally. To that end, after considering potential strategies, a mean reversion strategy with a beta-zero targeting overlay was chosen as a starting point.

Market research reports evidence that returns are negatively serially correlated.[2] The hypothesis that stock prices follow a random walk can be rejected[1] and it can be shown that there is significant autocorrelation in long-horizon returns.[2] If implemented properly, a mean reversion strategy can be used to generate alpha - a suggestion supported by Fama and French.

In addition to mean reversion, a mechanism that would minimize the portfolio's beta could enhance performance. Black, Jensen, and Scholes [BJS, 1972] and Miller and Scholes [1972] found that in the period from 1931 through 1965, low-beta stocks in the United States did better than the capital asset pricing model predicts, while high-beta stocks did worse. Several authors find that this pattern continued in subsequent years, at least through 1989. Fama and French extend it through 1990.[1] Extending the same idea to portfolios - that low-beta portfolios will outperform high-beta portfolios over long periods of time - could prove to be true.

Quantopian.com, a website that provides historical data and an online development environment, was used to code the algorithm in Python and run backtests.

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Current Methodology

The algorithm is outlined below:

Mean Reversion

1. After the market closes on Friday, scrub the week's trading data to find the stocks with the highest dollar volume liquidity. In this case, the top 3% of stocks by dollar volume in a one-day lookback.
2. Define the "low return group" as the set of the stocks with the bottom 3.6% of weekly returns among those in the high dollar volume group
3. Define the "high return group" as the set of the stocks with the highest 0.4% of weekly returns among those in the high dollar volume group
4. Buy the low return group and short the high return group on Monday at noon

Beta-Zero Targeting

5. Buy/short the index (SPY) and/or the 20-year treasury bond (TLT) to target a beta of 0 (relative to SPY), based on a one-year lookback

The variables here - shown in red above - were all datamined to produce the best results in backtests, so some more thought should be put into their exact values for a more robust solution. Going forward, perhaps a Monte Carlo analysis or some more research could aid in finding ideal parameters.

Initial Results



Fig. 1: Ten Year Backtest Results

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A ten-year backtest showed promising results. In a backtest from 1/1/2007 to 1/1/2017, the strategy produced a total return of 1303.51% against an S&P 500 total return of 93.16%, or an annualized return of 30.23% against a benchmark annualized return of 6.81%.

The algorithm began paper trading on 1/23/17, and its results are below in Fig. 2. While the sample size is limited, it is performing well out-of-sample.

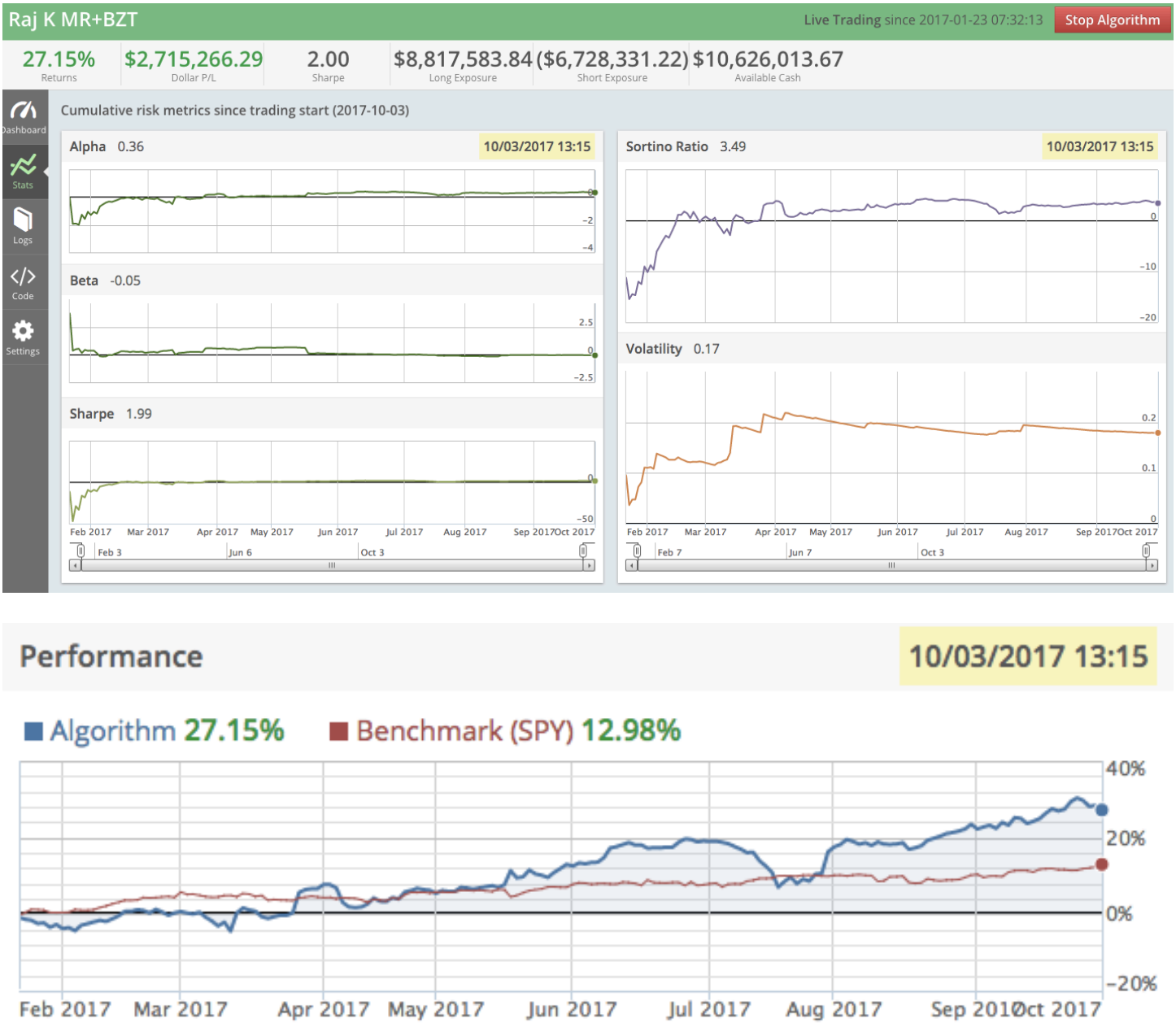


Fig. 2: Live Paper Trading Results, 1/23/17-10/3/17

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Next Steps and Analysis Iterations

- Optimization vs. Robustness

  The values of the algorithm's current parameters make it somewhat optimized, but its robustness can be improved. Its maximum drawdown since it began trading is -10.52%, which is far from ideal. A Monte Carlo analysis could be performed to ascertain the ideal parameters for the algorithm.

- Sector Concentration Considerations

  Because the strategy focuses solely on high dollar volume stocks, highly traded sectors are automatically favored. For example, healthcare and pharmaceutical stocks are often shorted the week after news of their successful drug trials. A check could be put into place to limit sector concentration to lower risk.

- Potential Role of Human Intervention

The fundamentals are ignored in purely technical strategies like this one, so mistakes are bound to be made without human intervention - for example, this strategy went long Equifax the Monday after their data breach was announced. Using the algorithm as a guide rather than a rulebook is necessary. A potential solution could involve producing the next week's proposed trades on Friday after the market closes and then doing research over the weekend to validate those trades before execution.

Academic References

1. Black, F. (1993). Beta and Return. *The Journal of Portfolio Management, 20*(1), 8-18.

2. Ceccheti, S.G., Lam, P., & Mark, N.C. (1990). Mean Reversion in Equilibrium Asset Prices.

   American Economic Review, 80(3), 398-418.

3. Poterba, J.M., & Summers, L.H. (1988). Mean Reversion in Stock Prices: Evidence and

   Implications. *Journal of Financial Economics, 22*(1), 27-59.

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Performance Update

Below are updated performance metrics as of 1/23/18, one year since the start of live trading.

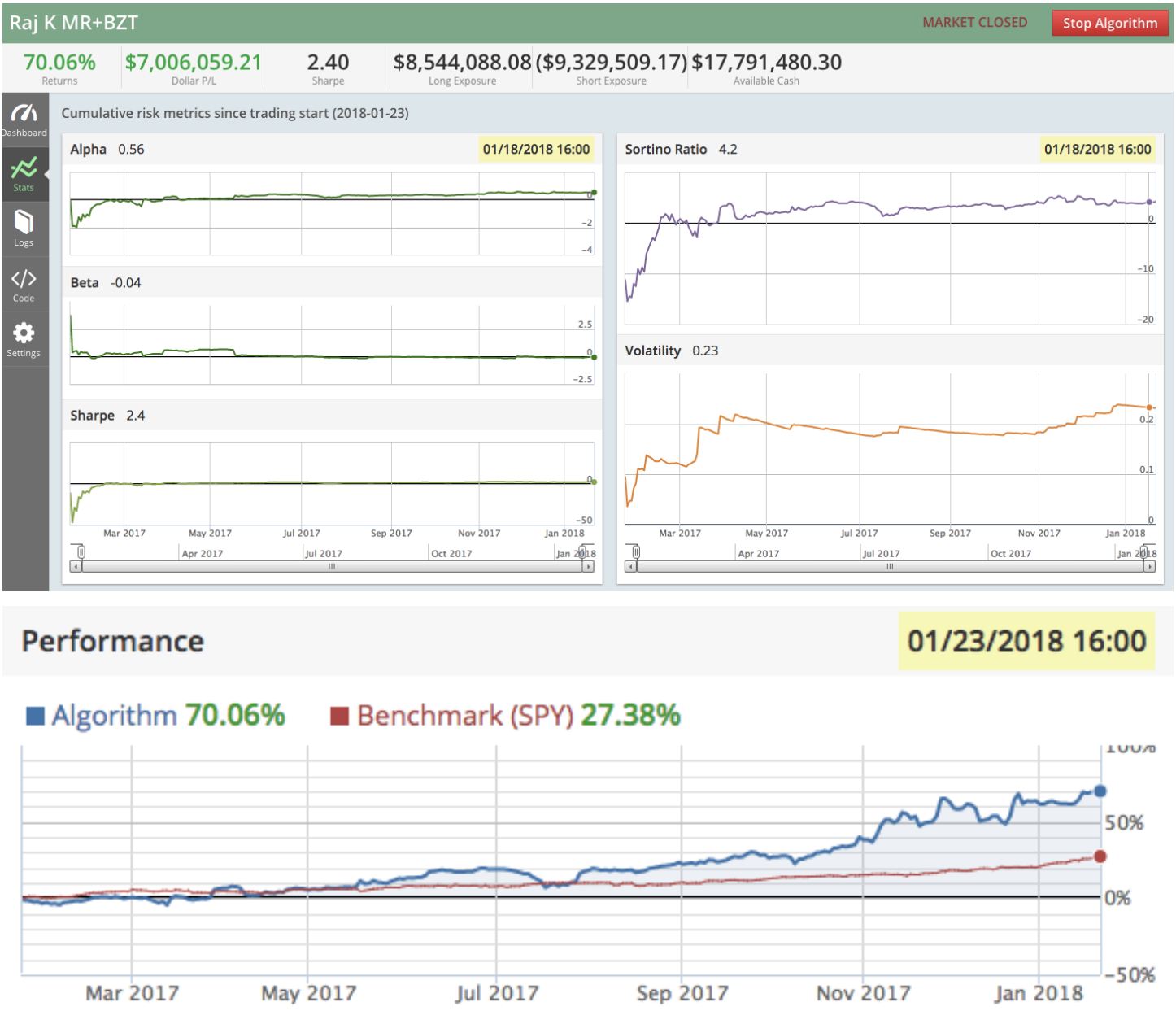


Fig. 3: Live Paper Trading Results, 1/23/17-1/23/18

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Corrections

Upon going over the code while attempting to refine the strategy, it was discovered that a one-day lookback was being used to define the high dollar volume group rather than a five-day lookback. This has been corrected in the methodology section on page 3, but because the intention was to use a five-day lookback rather than a one-day lookback, 10-year backtest results for the corrected algorithm are shown below and it remains to be seen how well the corrected algorithm performs while live paper trading.

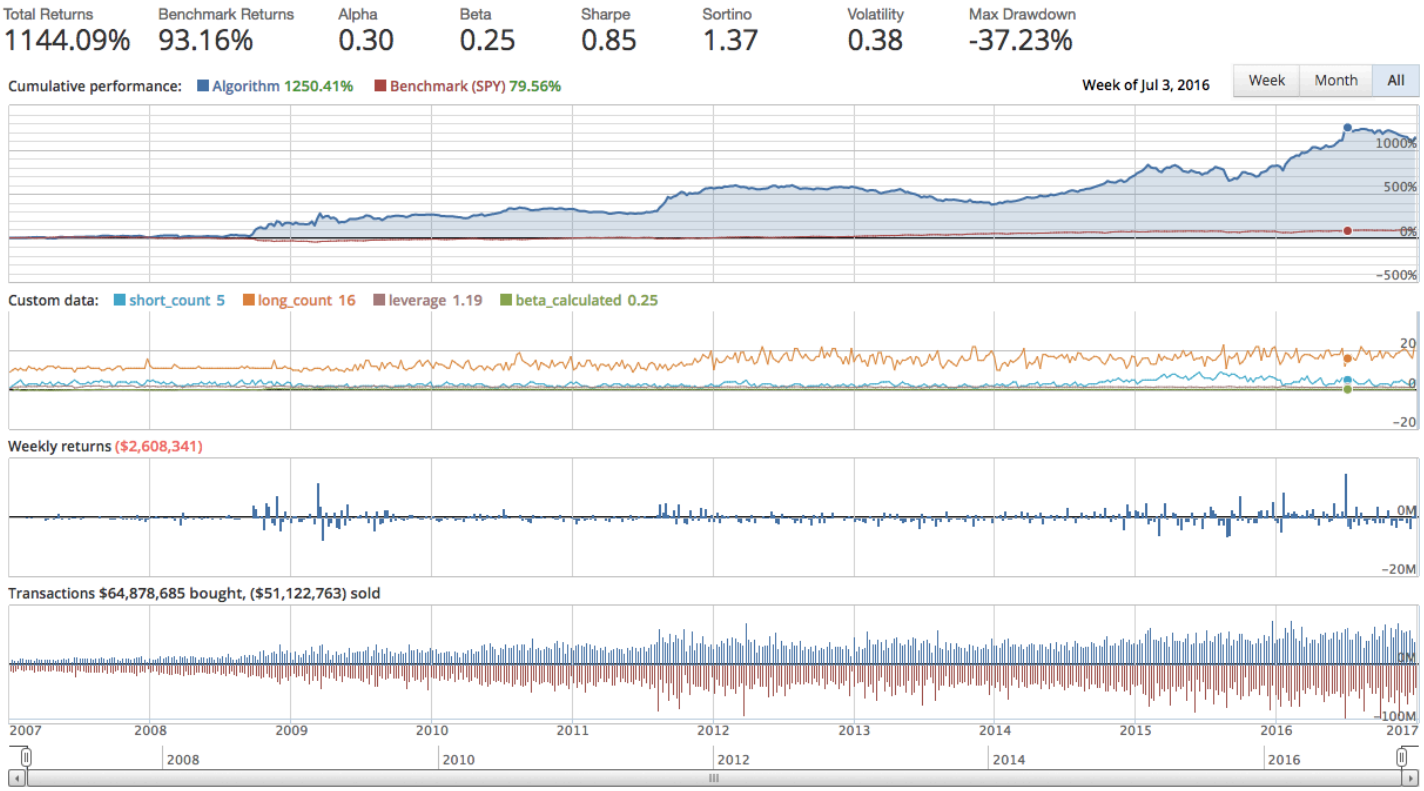


Fig. 4: Ten Year Backtest Results, Corrected Algorithm

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